

SmartClass Vision

Automated AI Face Recognition Attendance & Classroom Governance System

Repository: github.com/sandeepsagar18/Smart_Ai_class	Release: v0.2 Production
Platform: Windows / Desktop Application	Runtime: Python 3.10+ / CustomTkinter

1. Executive Overview & Problem Statement

SmartClass Vision is a computer vision solution designed to automate classroom attendance and eliminate proxy attendance. Traditional attendance workflows (manual roll calls, sign-in paper sheets, and RFID badge swipes) are prone to proxy marking, administrative burden, and human error. SmartClass Vision provides fast multi-face detection, biometric identification, anti-spoofing liveness verification, and role-based access control (RBAC).

2. Machine Learning & Deep Learning Architecture

The system incorporates a two-stage deep learning pipeline combining convolutional neural network (CNN) face detection with deep metric learning:

Module	Model / Algorithm	ML / DL Mechanism	Output
Face Detection	Ultralytics YOLOv8n	Single-stage Deep CNN object detector. Detects and tracks faces across varying lighting and classroom angles.	Bounding boxes [x1, y1, x2, y2] & cropped faces
Biometric Embedding	FaceNet512 (DeepFace)	Deep Metric Learning trained on Triplet Loss. Maps facial geometry into a dense 512-dimensional Euclidean hyperspace.	512-D continuous floating-point vector
Face Identification	Cosine Distance Metric	Computes Cosine Similarity: $\cos(u, v) = (u \cdot v) / (u \cdot v)$. Compares live embedding against saved student vectors. Threshold: < 0.25 .	Roll Number & Match Confidence %
Anti-Spoofing Guard	Laplacian Filter & Chroma Mask	Analyzes Laplacian frequency variance (detects paper blur / screen moire) and YCrCb/HSV skin locus balance (detects display glare).	Liveness score (Live Human vs Fake)

3. Functional Modules & Security Features

A. Role-Isolated Authentication Architecture:

- **Teacher Portal:** Contains Teacher Login and Create Teacher account. Keeps classroom operations clean and free of administrative controls.
- **Admin Portal:** Protected interface providing Admin Login and Create Admin account, gated by an institutional Master Authorization Key.
- **Brute-Force Lockout Defense:** Automatically locks an account after 5 consecutive failed password attempts.

- **Session Auto-Lock:** Automatically locks active classroom terminals after 10 minutes of inactivity.

B. Interactive Student Registration & Profile Directory:

- **Multi-Angle Capture:** Directs students to look straight, turn left, turn right, and tilt up/down to store a 5-shot training dataset.
- **Student Profile Modal:** Searching by Roll Number opens a verified Student Profile Card showing the captured face photo, roll number, name, degree, branch, section, and enrollment status.

C. Batch Class Attendance Engine:

- **Section Isolation:** Restricts scanning strictly to students enrolled in the teacher's selected degree, year, branch, and section.
- **Temporal Multi-Shot Verification:** Validates identity over multiple frames to avoid false-positive attendance marks.
- **Cryptographic HMAC-SHA256 Signatures:** Every generated attendance sheet is signed with SHA-256 HMAC checksums to ensure file integrity.

4. Technology Stack

Layer	Technology	Role
GUI Framework	CustomTkinter, Tkinter	Desktop interface with dark mode and kiosk layout
Computer Vision	OpenCV (cv2), Pillow (PIL)	Real-time webcam stream, DirectShow capture, face crops
Deep Learning	PyTorch, TensorFlow, DeepFace, YOLOv8	Face detection inference and FaceNet512 512-D embeddings
Database	SQLite3, Pandas, Pickle	Master database, attendance exports, binary embedding storage
Security	HMAC-SHA256, Salted Hashes	Argon2/PBKDF2 style hashing, attendance file tamper detection

5. Default Credentials & Quick Reference

Item	Value	Description
Default Admin Account	ID: ADMIN01 Password: admin123	Full admin command portal, faculty allotments, diagnostics
Admin Master Auth Key	SmartClass@Admin#2026	Required for creating administrator accounts
Quick Launch Script	run.bat or python gui.py	Starts the desktop application within the virtual environment